

User guide

1. General information

The PAH HBM tool supports the interpretation and decision-making process regarding 1-hydroxypyrene (1-OHP) concentrations, by providing relevant contextual and analytical information.

The results generated by the tool are valid only for the exposure scenario and parameters defined by the user, within the limits and uncertainties of the available data (e.g., literature data) and methods applied. The responsibility for selecting the most relevant total pyrene/benzo[a]pyrene (PyrT/BaP) ratio, parameters, and exposure scenario rests with the user.

The interface is organized into a blue box panel (parameters) and nine tabs. *Figure 1* presents a screenshot of the tool. All tables and figures are interactive.

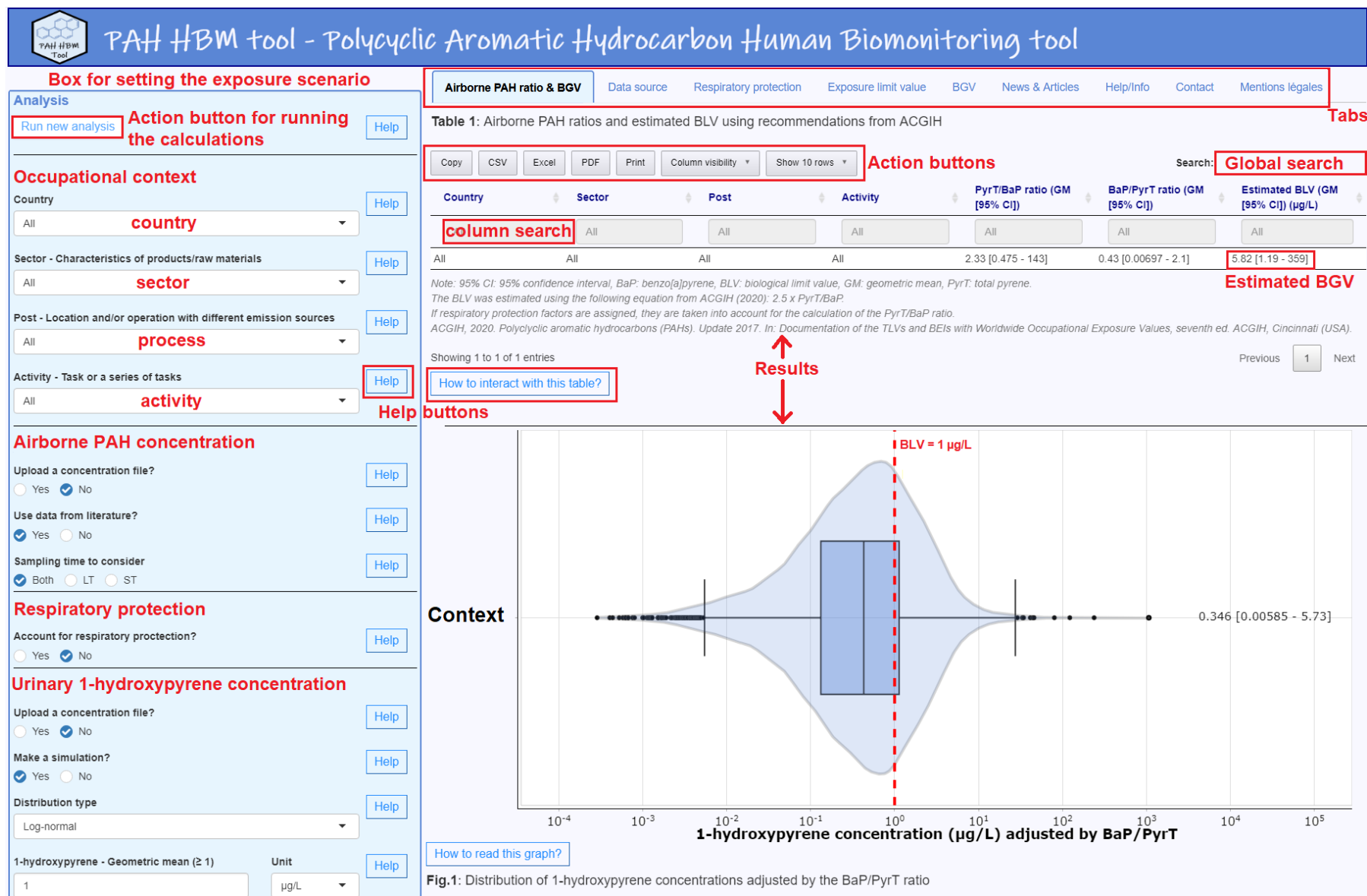


Figure 1: Screenshot of the PAH HBM tool

2. Help buttons and interactivity

[Help](#) Help buttons are available throughout the tool. They provide guidance for setting parameters, reading graphs, and interacting with tables and figures.

2.1 Table interactivity

[How to interact with this table?](#) A help button titled “*How to interact with this table?*” is located below each table.

Above each table (top-left), seven action buttons are displayed (*Figure 2*):

- **Copy**: Copy the table to the clipboard.
- **CSV, Excel, PDF**: Export the table in the chosen format.
- **Print**: Print the table.
- **Column visibility**: Select which columns to display.
- **Show 10 rows**: Choose how many rows to display.

⚠ Important: To export or print an entire table, you must first set the display option to “*Show all rows*”, then use one of the export/print buttons.

Search and sorting features:

- A global search bar (top-right of the table) filters across all searchable columns.
- Individual column search bars (below each header) allow column-specific queries.
- Queries are not case-sensitive, update in real time, and are flexible regarding word order.
- Use the X icon to cancel a search.
- Use the sorting arrow on column headers to sort alphabetically or numerically.
- Columns can be reordered by dragging their headers.

Additional notes:

- References and external links are underlined in blue; clicking them opens the source.
- Empty cells correspond to non-reported values.

Table 1: Airborne PAH ratios and estimated BGV using recommendations from ACGIH

Copy	CSV	Excel	PDF	Print	Column visibility ▾	Show 10 rows ▾	Search:
Country	Sector	Process	Activity	PyrT/BaP ratio (GM [95% CI])	BaP/PyrT ratio (GM [95% CI])	Estimated BGV (GM [95% CI]) (µg/L)	
All	All	All	All	All	All	All	
All	All	All	All	2.58 [0.445 - 113]	0.387 [0.00888 - 2.25]	6.45 [1.11 - 282]	

Note: 95% CI: 95% confidence interval, BaP: benzo[a]pyrene, BGV: biological guidance value, GM: geometric mean, PyrT: total pyrene.
The BGV was estimated using the following equation from ACGIH (2020): $2.5 \times \text{PyrT}/\text{BaP}$.
If respiratory protection factors are assigned, they are taken into account for the calculation of the PyrT/BaP ratio.
ACGIH, 2020. Polycyclic aromatic hydrocarbons (PAHs). Update 2017. In: Documentation of the TLVs and BEIs with Worldwide Occupational Exposure Values, seventh ed. ACGIH, Cincinnati (USA).

Showing 1 to 1 of 1 entries

[How to interact with this table?](#) Previous **1** Next

Figure 2 : Example of a table

2.2 Figure interactivity

[How to read this graph?](#) A help button titled “How to read this graph?” is located below each figure.

When hovering over a figure:

- A tooltip displays details about the selected data point.
- Icons appear in the top-right corner of the graph (*Figure 3*), providing options such as download (camera icon) and zoom.

- If a legend is present, data series can be toggled on/off by clicking on them

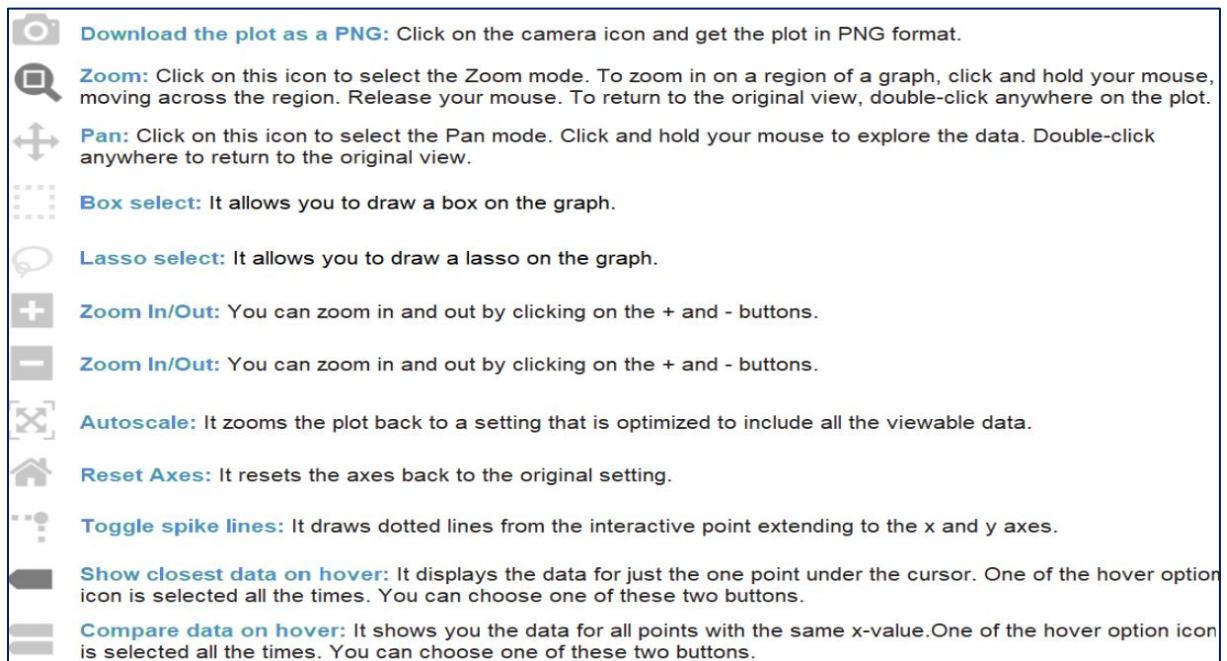


Figure 3: Graph interactivity functionalities

3. Blue box panel (parameter settings)

The **blue box panel** (left side of the interface) is used to define the exposure scenario (*Figure 4*).

The screenshot shows a web-based parameter settings interface for exposure analysis. It is organized into several sections, each with a title and a 'Help' button. The 'Analysis' section at the top contains a 'Run new analysis' button. The 'Occupational context' section includes dropdown menus for 'Country', 'Sector - Characteristics of products/raw materials', 'Process - Location and/or operation with different emission sources', 'Activity - Task or a series of tasks', and 'Data source'. The 'Airborne PAH Concentrations' section features radio buttons for 'Upload a concentration file?' (No is selected), 'Use data from the literature review and/or E-HAP?' (Yes is selected), and 'Sampling time to consider' (Long term (≥ 2h) is selected). The 'Respiratory Protection' section has a radio button for 'Account for respiratory protection?' (No is selected). The 'Urinary 1-hydroxypyrene Concentrations' section has a radio button for 'Upload a concentration file?' (No is selected). The 'Biological Exposure Guidance Value (BGV)' section has a radio button for 'Use the estimated indicative BGV for 1-hydroxypyrene?' (Yes is selected).

Analysis
Run new analysis Help

Occupational context
Country All Help
Sector - Characteristics of products/raw materials All Help
Process - Location and/or operation with different emission sources All Help
Activity - Task or a series of tasks All Help
Data source Consider everything Help

Airborne PAH Concentrations
Upload a concentration file? ☐ Yes ☒ No Help
Use data from the literature review and/or E-HAP? ☒ Yes ☐ No Help
Sampling time to consider ☐ Both ☒ Long term (≥ 2h) ☐ Short term (< 2h) Help

Respiratory Protection
Account for respiratory protection? ☐ Yes ☒ No Help

Urinary 1-hydroxypyrene Concentrations
Upload a concentration file? ☐ Yes ☒ No Help

Biological Exposure Guidance Value (BGV)
Use the estimated indicative BGV for 1-hydroxypyrene? ☒ Yes ☐ No Help

Figure 4: Blue box panel

3.1 Run new analysis

The **Run analysis** button applies any changes made in the parameter box. Results will update only after clicking this button. ⚠ Depending on the number of simulation runs, internet connection, and computer performance, generating results may take from a few seconds to several minutes.

3.2 Occupation context

Users begin by defining an occupational context:

- Country
- Similar exposure group (SEG): sector, process/post, activity

To view the full list of available options, click the dropdown menus (white area or black arrow).

Definitions:

- **Sector:** Industry or product/raw material type.
- **Process:** Location or operation with distinct emission sources.
- **Activity:** Task or series of tasks.

For details, see *Petit et al. 2017*.

Data sources

Users may choose between:

- Published literature data,
- The Exporisq-HAP dataset,
- Or both (if available).

⚠ Only one option can be selected at a time.

⚠ If you upload your own concentration file, this section is ignored.

For details on data availability, see the **Data Source** tab.

3.3 Airborne PAH concentration settings

Users can either:

- Upload a file with measured airborne PAH concentrations, or
- Use default/simulated data (if no file is uploaded).

File upload requirements (Figure 5):

- Format: 7-column .csv (semicolon) or .txt.
- Columns (in strict order):
 1. Country
 2. Sector
 3. Process/Post/Job title
 4. Activity
 5. BaP concentration ($\mu\text{g}/\text{m}^3$)
 6. PyrP concentration ($\mu\text{g}/\text{m}^3$)
 7. PyrG concentration ($\mu\text{g}/\text{m}^3$)

⚠ **Rules:**

- Columns must be ordered exactly as above.
- Last 3 columns: numeric, positive, no missing values.
- At least 1 row required.
- File/path name must not contain special characters (é, ï, ù, etc.).
- All uploaded files (e.g., PAH concentrations, APF, urinary 1-OHP) must have the same item order in column 1.

⚠ After uploading, click **Run new analysis** to process the file.

⚠ Uploaded files are not stored on servers; data privacy and anonymity are guaranteed.

△ To remove a file, refresh the page.

Country	Sector	Post	Activity	BaP	PyrP	PyrG
All	All	All	All	10	1	4
All	All	All	All	1	2	5
All	All	All	All	0.5	3	6
All	All	All	All	6	4	34
All	All	All	All	3	5	63
France	Aluminum production	Paste tower	Maintenance	0.1	0.1	10
France	Aluminum production	Paste tower	Maintenance	0.2	0.2	10
France	Aluminum production	Paste tower	Maintenance	0.3	0.3	11
France	Aluminum production	Paste tower	Maintenance	0.1	0.4	11
Germany	Foundry	Aluminum foundry	Furnace	0.1	0.1	10
Germany	Foundry	Aluminum foundry	Furnace	0.2	0.2	10
Germany	Foundry	Aluminum foundry	Furnace	0.3	0.3	11
Germany	Foundry	Aluminum foundry	Furnace	0.1	0.4	11
Germany	Foundry	Aluminum foundry	Furnace	0.32	0.7	13
Hungary	Silicon production	Composite electrode	Paste loading	100	20	3

Figure 5: Example of an accepted airborne PAH concentration file structure

Simulation (if no file is uploaded and the user do not wish to use data from the literature/Exporisq-HAP):

- Choose a distribution type (default: log-normal).
- Define parameters (mean, deviation).
- Select number of samples (≥ 5 ; default = 101).
- Select number of simulation runs (default = 1,000).

△ Higher run numbers increase computation time.

Using data from the literature/Exporisq-HAP:

- Choose to use literature, Exporisq-HAP or both data.
- Select sampling duration (long-term ≥ 2 h, short-term < 2 h, or all).

△ For cancer risk assessment and BGV estimation, only long-term sampling should be used.

3.4 Respiratory protection

- Users can include the effect of respiratory protective equipment (RPE) in their analysis.
- Efficiency is applied via an assigned protection factor (APF).

$$_{adj}1OHP = 1OHP \times \left[\frac{wBaP}{wtPyr} \right] = 1OHP \times \left[\frac{wBaP}{wgPyr + wpPyr} \right]$$

Note: adj: adjusted, wBaP: benzo[a]pyrene concentration weighted by an APF, wtPyr: total pyrene weighted by an APF, wgPyr: gaseous pyrene weighted by an APF, wpPyr: particulate pyrene weighted by an APF, 1-OHP: urinary 1-hydroxypyrene

- Examples: A particulate mask (APF = 10 for particulate PAHs, APF = 1 for gaseous PAHs) reduces particulate PAHs by 90% while leaving gaseous PAHs unchanged.

Assigned protection factor used for the weighted level calculation depending on the type of mask worn.				
Type of mask worn	APF for pPAHs	APF for gPAHs	pPAH levels	gPAH levels
No mask	1	1	Unchanged	Unchanged
Gas mask	1	10	Unchanged	Divided by 10
Particulate mask	10	1	Divided by 10	Unchanged
Combined mask	10	10	Divided by 10	Divided by 10
SAR	250	250	Divided by 250	Divided by 250

Note: APF: assigned protection factor, gPAHs: gaseous PAHs, pPAHs: particulate PAHs, SAR: supplied-air respirator.

- APF values depend on mask type (see the **Respiratory Assigned Protection Factors** tab).

For methodology, see Valière et al. 2022.

3.5 1-OHP concentration settings

Users can either:

- Upload a file with measured urinary 1-OHP concentrations (default), or
- Run a simulation (if no file is uploaded).

File upload requirements (Figure 6):

- Format: 5-column .csv (semicolon) or .txt.
- Columns (strict order):
 1. Country
 2. Sector
 3. Process/Post/Job title
 4. Activity
 5. 1-OHP concentration (µg/L)

Rules:

- Last column: numeric, positive, no missing values.
- At least 1 row required.
- File/path name must not contain special characters.
- Occupational context in the 1-OHP file must match that specified in the tool (if no PAH file uploaded). Otherwise, the tool may crash.

⚠ After uploading, click **Run new analysis**.

⚠ Files are not stored; privacy is guaranteed.

Country	Sector	Post	Activity	1-OHP
All	All	All	All	10
All	All	All	All	1
All	All	All	All	0.50
All	All	All	All	6
All	All	All	All	3
France	Aluminum production	Paste tower	Maintenance	0.12
France	Aluminum production	Paste tower	Maintenance	0.21
France	Aluminum production	Paste tower	Maintenance	0.37
France	Aluminum production	Paste tower	Maintenance	0.10
Germany	Foundry	Aluminum foundry	Furnace	0.25
Germany	Foundry	Aluminum foundry	Furnace	0.46
Germany	Foundry	Aluminum foundry	Furnace	0.54
Germany	Foundry	Aluminum foundry	Furnace	0.28
Germany	Foundry	Aluminum foundry	Furnace	0.72
Hungary	Silicon production	Composite electrode	Paste loading	4

Figure 6: Example of an accepted 1-OHP urinary concentration file structure

Simulation (if no file is uploaded):

- Choose a distribution type (default: log-normal).
- Define parameters (geometric/arithmic mean, standard deviation).
- Select number of samples (≥ 5 ; default = 101).
- Select number of runs (default = 1,000).

⚠ Higher run numbers increase computation time.

3.6 Biological Exposure Guidance Value (BGV)

Users can either:

- Use the indicative BGV estimated according to ACGIH recommendations (default), or
- Define their own BGV.

If a user-defined BGV is entered, results will display both the indicative and user-specified values.

⚠ This option is only considered if urinary 1-OHP data are available (via upload or simulation).

4. Tabs

Nine tabs are displayed at the top of the interface. The active tab is highlighted (dark blue background, bold font).

4.1 Airborne PAH ratio & BLV tab

This tab displays:

- The estimated BGV (ACGIH method), based on the chosen scenario.
- Distribution of 1-OHP concentrations (adjusted by PyrT/BaP ratio).
- Probability of exceeding the estimated BGV.

Outputs:

- *Table 1*: Airborne PAH ratios and estimated BGVs.
- *Figure 1*: Distribution of urinary 1-OHP by occupational context, shown as boxplots + violin plots. Geometric mean and 95% CI are displayed (Figure 7 below).
- *Table 2*: Summary of distribution parameters and BGV exceedance probabilities.

△ Hovering over *Figure 1* shows log-transformed values. To recover non-log values, compute $10^{(-\text{displayed value})}$.

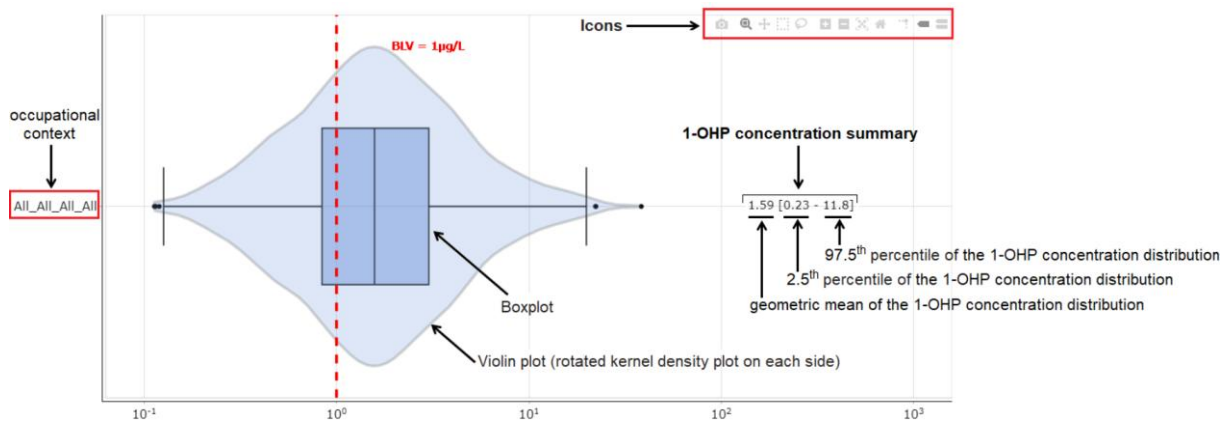


Figure 7: How to read the 1-OHP concentration distribution graph

If both 1-OHP and airborne PAH files (≥ 3 contexts) are uploaded, a matrix of degrees of agreement is generated (Figure 8). Each cell shows % overlap between distributions (0% = no overlap, 100% = complete overlap). See *Petit and Bicout, 2022*.

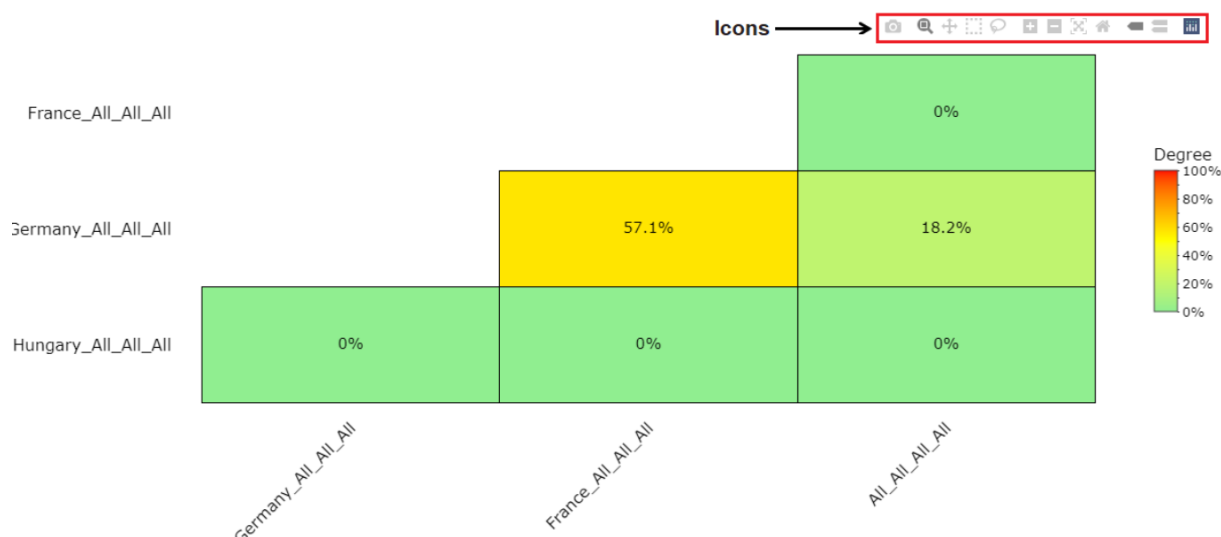


Figure 8: Example of a matrix of degrees of agreement between 1-OHP concentration (adjusted by the BaP/PyrT ratio) distributions of 4 occupational contexts

4.2 Other tabs

- **“Data source”**: Inventory of PyrT/BaP ratios from literature and Exporisq-HAP dataset.
- **“Respiratory assigned protection factors”**: Table of APF values from literature.
- **“Exposure limits values”**: Table of airborne BaP and urinary 1-OHP limits.
- **“BGV”**: Recommendations on choosing PyrT/BaP ratios by sector, process, or activity (see Clauzel et al. 2024).
- **“News & Articles”**: Publications describing or citing the PAH HBM tool.
- **“Help/Info”**: General tool information, strengths/limitations, interactivity guidance, privacy/legal details, and bug reporting.
- **“Contact”**: Contact information.
- **“Mentions légales”**: Legal notices (in French).

5. References

Clauzel A, Persoons R, Maître A, Balducci F, Petit P. Review of environmental airborne pyrene/benzo[a]pyrene levels from industrial emissions for the improvement of 1-hydroxypyrene biomonitoring interpretation. *J Toxicol Environ Health B Crit Rev.* 2024;27(5-6):212-232. <https://doi.org/10.1080/10937404.2024.2362632>.

Petit P, Bicout DJ, Persoons R, Bonnetterre V, Barbeau D, Maître A. Constructing a Database of Similar Exposure Groups: The Application of the Exporisq-HAP Database from 1995 to 2015. *Ann Work Expo Health.* 2017;61(4):440-456. <https://doi.org/10.1093/annweh/wxx017>.

Petit P, Bicout DJ. Health risk assessment with multiple reference indices. *Sci Total Environ.* 2022;804:149971. <https://doi.org/10.1016/j.scitotenv.2021.149971>.

Valière M, Petit P, Persoons R, Demeilliers C, Maître A. Consistency between air and biological monitoring for assessing polycyclic aromatic hydrocarbon exposure and cancer risk of workers. *Environ Res.* 2022;207:112268. <https://doi.org/10.1016/j.envres.2021.112268>.